14–DAMAGE CONTROL

14.1 INTRODUCTION

The survival of a ship at sea is very much dependent on its ability to:

- Right itself when heeled
- Remain stable both when intact or when flooded
- Withstand fire damage

14.2 LIMITS OF SURVIVABILITY/MARGIN LINE

Remaining stable after flooding is dependent on the amount of water a ship will allow to enter. This is achieved through subdividing the hull into compartments, such that in event of damage, a particular section can be sealed off to maintain water-tight integrity and allow the ship to continue its mission without being affected.

Now, it is generally accepted that in any condition of flooding, a vessel will survive if the resulting waterline at which she floats does not go beyond the FREEBOARD DECK - the uppermost complete deck exposed to weather and sea, which has permanent means of closing all openings in the weather part, and to which all bulkheads terminate.

However, to provide a higher margin of safety, an imaginary line drawn 76 mm below and parallel to the bulkhead deck at sides is rather used as the limiting waterline for survival. This line is referred to as the **MARGIN LINE (ML)**.

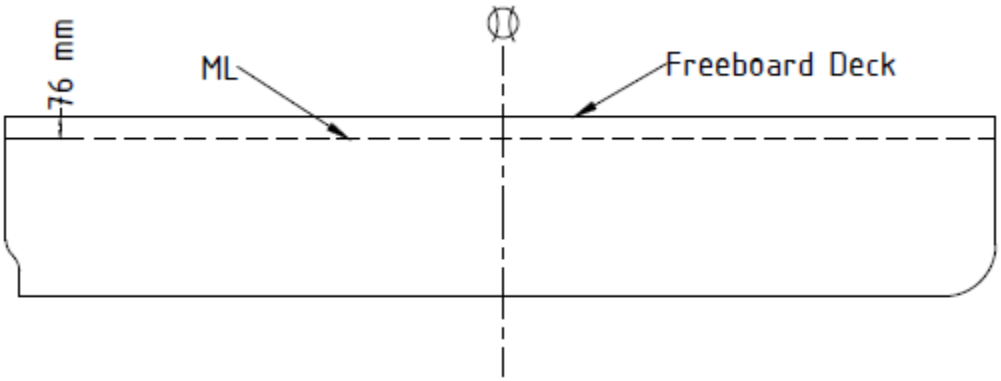


Figure 14.1 – Margin Line (ML)

Determining how a vessel must be subdivided to meet above objective may be approached in two ways:

1. Floodable Length and Factor of Subdivision Method
2. Probabilistic Damage Assessment Method

14.3 FLOODABLE LENGTH METHOD

14.3.1 - Floodable Length Calculations

Consider a vessel of volume ∇ to be initially intact and in static equilibrium on even keel at a waterline (w_0l_0) as in Figure 14.2.

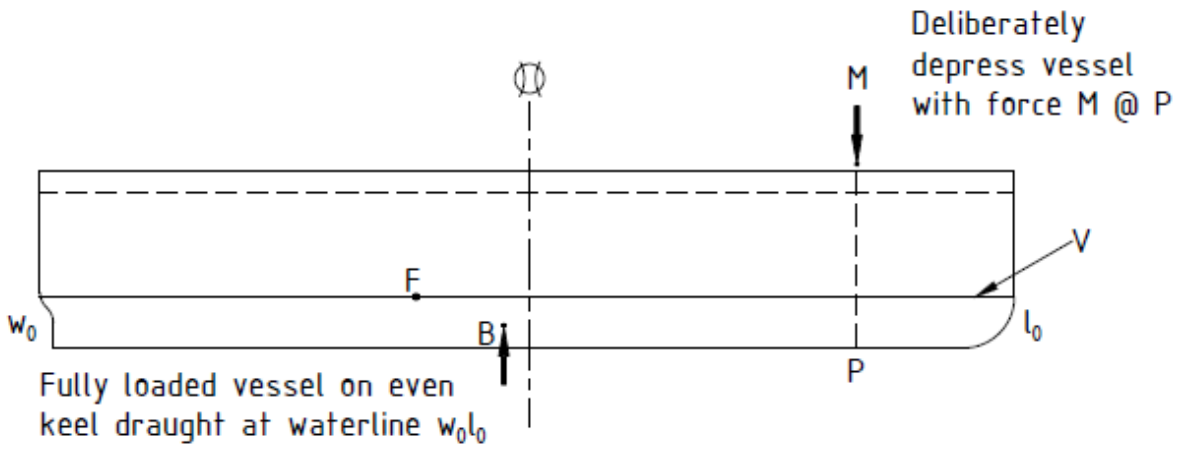


Figure 14.2

By simulation, deliberately apply a downwards force M to depress the vessel to waterline w1l1 to coincide with the ML.

The vessel will at this stage trim and volume of displacement will increase to ∇_1 as in Figure 14.3.

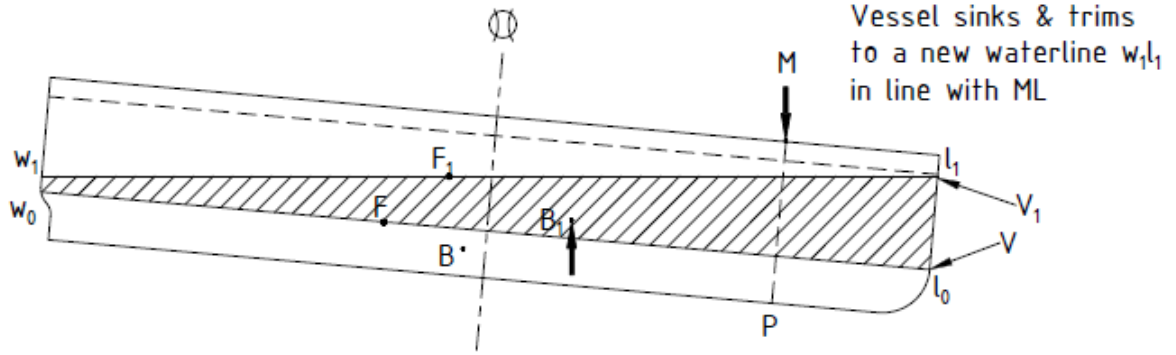


Figure 14.3

Under this new condition:

$$Increase\ in\ Volume = (\nabla_1 - \nabla)$$

Convert this increase to an equivalent rectangular volume as in Figure 14.4.

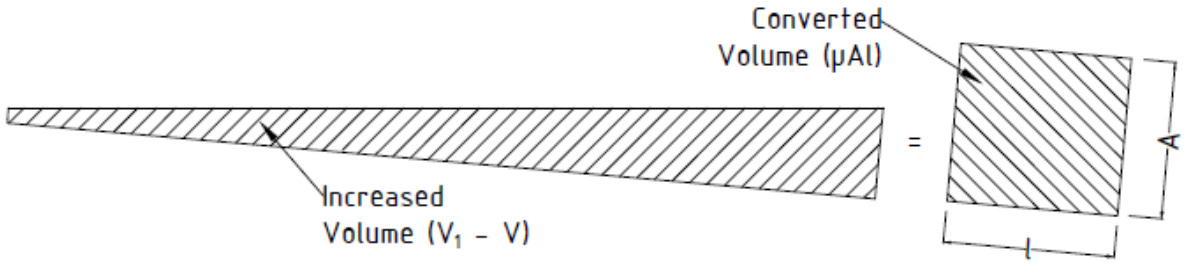


Figure 14.4

This will then mean that the:

$$Increase\ in\ Volume = Converted\ Volume$$

If therefore this equivalent volume can be carved out of the ship as a compartment at position P as in Figure 14.5, then in event of flooding, the vessel will be expected to survive up to the ML.

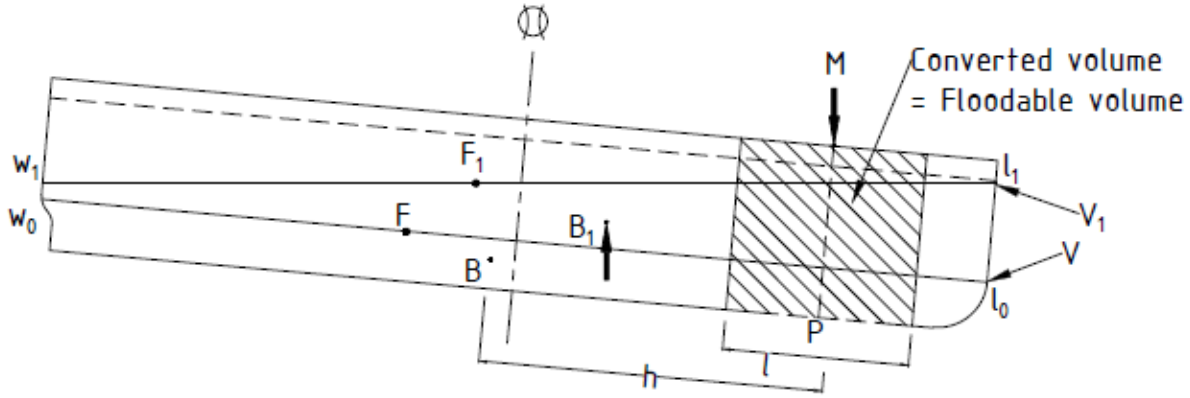


Figure 14.5 – Floodable Volume

This compartment is referred to as the **FLOODABLE VOLUME**.

If (A) represents the mean cross-sectional area of the volume, and (l) its length then:

$$Floodable\ volume = Increase\ in\ volume$$

$$lA = \nabla_1 - \nabla$$

And:

$$l = \frac{\nabla_1 - \nabla}{A}$$

This length is referred to as the **FLOODABLE LENGTH (FL)**. It is therefore the maximum length of a ship at an assumed point of damage that can be flooded without submerging the margin line.

With reference to Figure 14.5, let (h) be the distance of the centroid of the floodable volume from the initial centre of buoyancy (B) of the vessel.

By taking moments of volume about B longitudinally, and noting that the *increase in volume equals the floodable volume*, then:

$$(Initial\ Volume \times 0) + (Increased\ Volume \times h) = (Final\ Volume\ \times BB_1)$$

$$(\nabla_1 - \nabla)h = \nabla_1 \times BB_1$$

$$h = \frac{\nabla_1 \times BB_1}{\nabla_1 - \nabla}$$

14.3.2 – Floodable Length Curves

By repeating this simulation at a number of positions along the length of the vessel from AP to FP, a Curve of FL is obtained which has a typical shape as in Figure 14.6. It is the curve that, at every point in the vessel's length, has an ordinate representing the length of the ship that with the vessel being upright, may be flooded without the margin line being submerged.

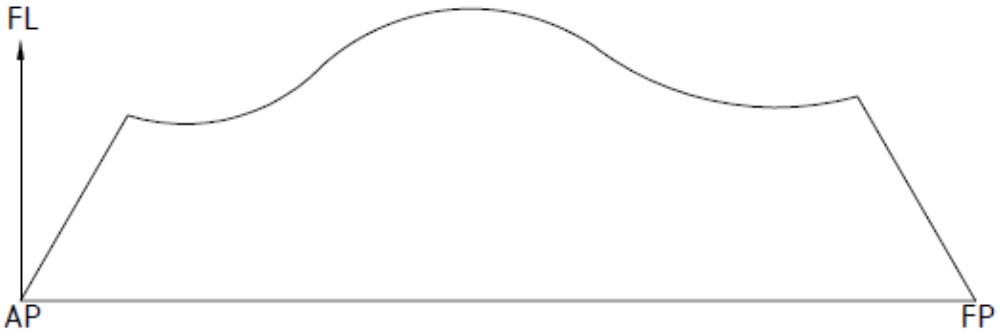


Figure 14.6 – Curve of Floodable Length

Now, machinery spaces in ships have different permeability from that of spaces forward and aft of it. Therefore, by further taking this into consideration, the FL calculation will modify to:

$$\mu l A = \nabla_1 - \nabla$$

From which:

$$l = \frac{\nabla_1 - \nabla}{\mu A}$$

By applying the SOLAS, Chapter II-1 (Reg. 5) standards of permeability stated below, the FL curves will then be modified as in Figures 14.7 and 14.8.

For machinery spaces:

$$\mu = 85 + 10\left(\frac{a - c}{v}\right)$$

Where:

a = volume of passenger spaces below the ML within the machinery space
c = volume of tween deck spaces below the ML within space concerned
v = whole volume of machinery space below the ML

For spaces forward of or abaft the machinery space:

$$\mu = 63 + 35\left(\frac{a}{v}\right)$$

Where:

a = volume of passenger spaces below the ML fore & aft of machinery space

v = whole volume of ship below the ML forward of or abaft the machinery space

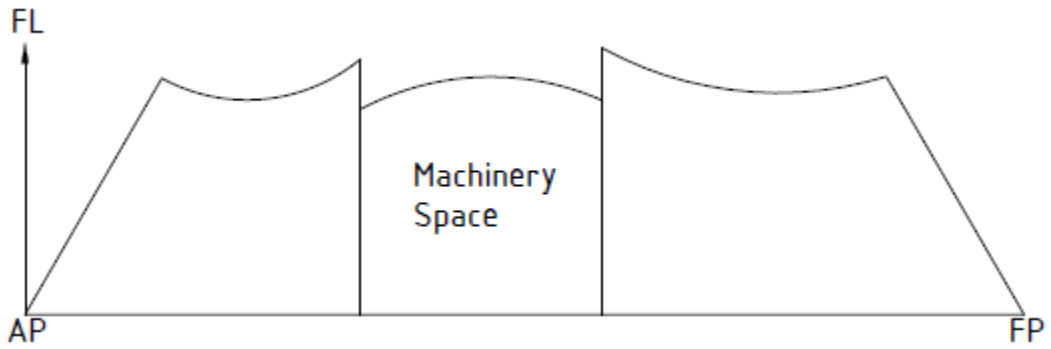


Figure 14.7 – Curve of Floodable Length (Machinery Amidship)

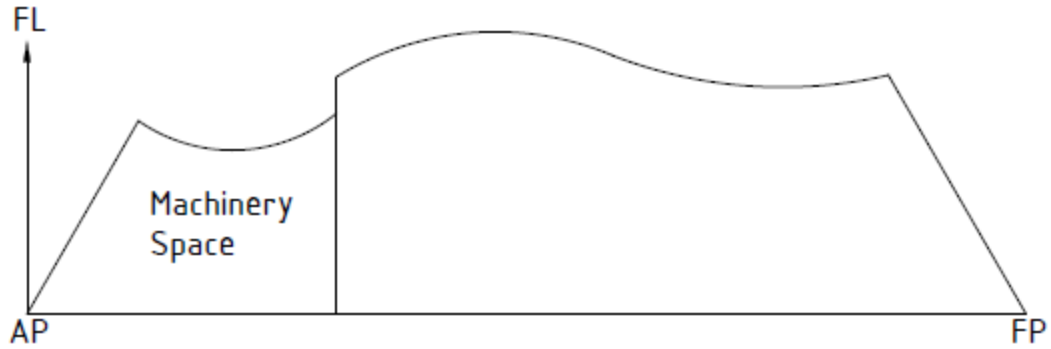


Figure 14.8 – Curve of Floodable Length (Machinery Aft)

To determine the bulkheads that define a compartment at any point, proceed as illustrated in Figure 14.9:

- Pick the FL
- Set the length horizontally on the AP-FP scale making sure its midpoint coincides with the desired position to form an isosceles triangle with slopes equal to 2:1, or having base angles equal to $\tan^{-1} 2$ degrees
- The endpoints of the horizontal length then mark the positions where the bulkheads will be placed

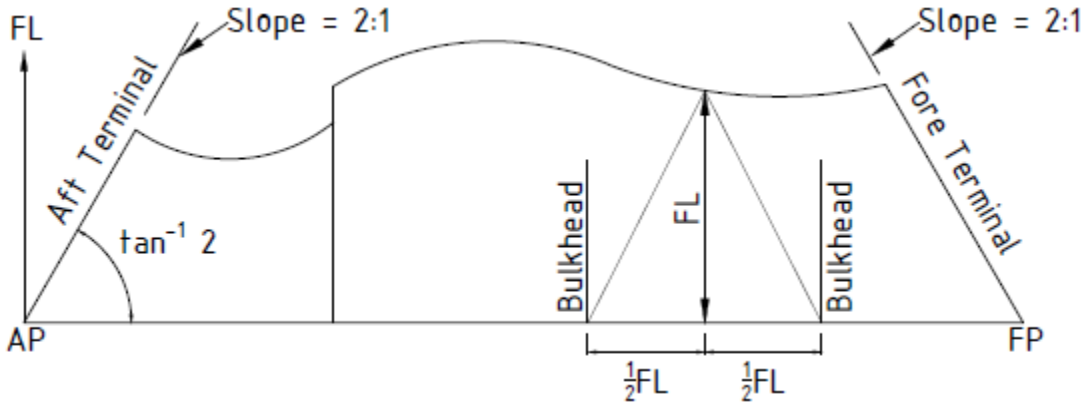


Figure 14.8 – Defining the Position of Bulkheads

14.3.3 – Permissible Length Curves

Regulations require that a ship should be able to float with any one compartment open to sea as the minimum. However, as lengths increase and due to varying nature of service, a higher standard of subdivision is required. This is achieved by the introduction of **Permissible Length (PL)**. It is some fraction of the FL and calculated by multiplying it with a **Factor of Subdivision (F_S)** as follows to obtain the typical curve shown in Figure 14.9.

$$PL = FL \times F_S$$

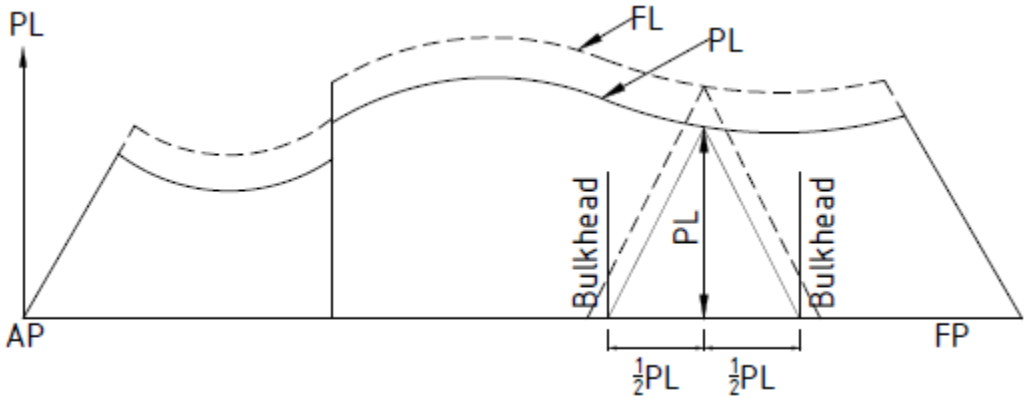


Figure 14.9 – Curve of Permissible Length

Broadly, the factor of subdivision is the inverse of the factor of subdivision and ensures that one, two or three compartments can be flooded before the margin line is immersed. This leads to what is referred to as **Compartment Standard**, or called ***one-, two-* or *three-compartment standard ships***.

In general terms the factor of subdivision decreases with length of ship and is lower for passenger ships than cargo ships and is determined by the formula:

$$\mathrm{F_S} = \mathrm{A} - \{(\mathrm{A} - \mathrm{B})(\mathrm{C_S} - 23)\} \times 100$$

Where:

$$\mathrm{A} = \frac{58.2}{(\mathrm{L} - 60)} + 0.18$$

$$\mathrm{B} = \frac{30.3}{(\mathrm{L} - 42)} + 0.18$$

It depends upon the length of a ship and a **Criterion of Service Numeral** or simply **Criterion Numeral (C_S)** which is calculated by:

$$C_s = 72\left(\frac{M + 2P}{V}\right)$$

Where:

M = Total volume of machinery spaces below the ML

P = Total volume of passenger space and crew space below the ML

V = Total volume of ship from keel to the ML

Criterion Numeral decreases in a regular and continuous manner with respect to ship length and nature of service from 123 for cargo ships to 23 for tankers.

14.3.4 – Checking Compliance with Damage Stability Requirements
(*By Capt. Rajeev Jassal*)

To check if a ship would comply with the damage stability requirement, the floodable length curve is superimposed on the ship's plan as follows:

One-Compartment Standard
Starting with one compartment at a time, the length of the assumed damaged compartment is plotted vertically at the centre of the compartment.

If this length is below the floodable length curve, then as in Figure 14.10 the compartment complies with the requirements of one compartment standard.

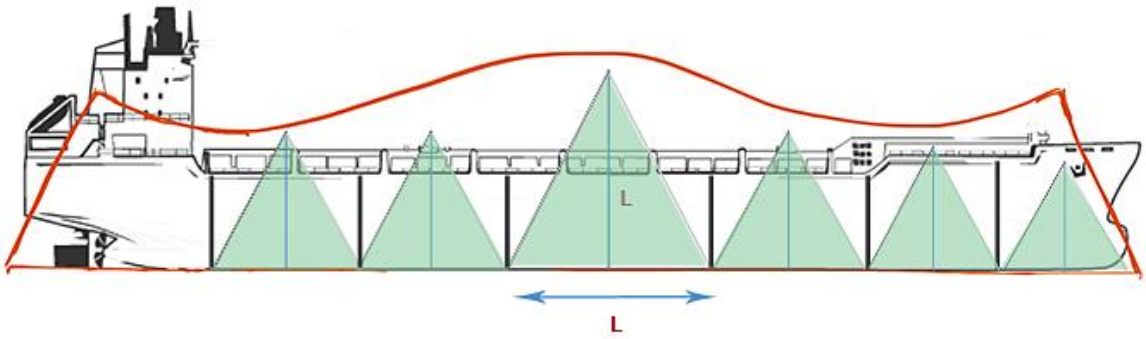


Figure 14.10 – Compliance with One Compartment Standard

By repeating the process for other compartments, it can be seen that all the length triangles are within the floodable length curve of the ship. This means that this ship complies with one compartment standards of the damage stability.

If, however, flooding of two compartments is assumed, it is clear from Figure 14.11, that the damage stability compliance for two compartment standard is not feasible.

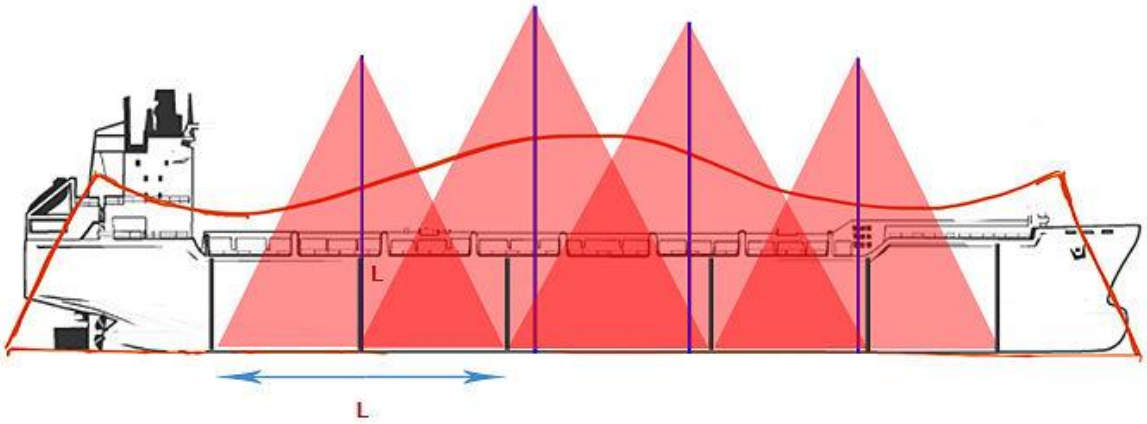


Figure 14.11 – Non-compliance with One Compartment Standard

Two-Compartment Standard

This time around, by assuming flooding of any two adjacent compartments, compliance with two compartment standards requirements will be met by introducing more bulkheads for the same ship length as in Figure 14.12.

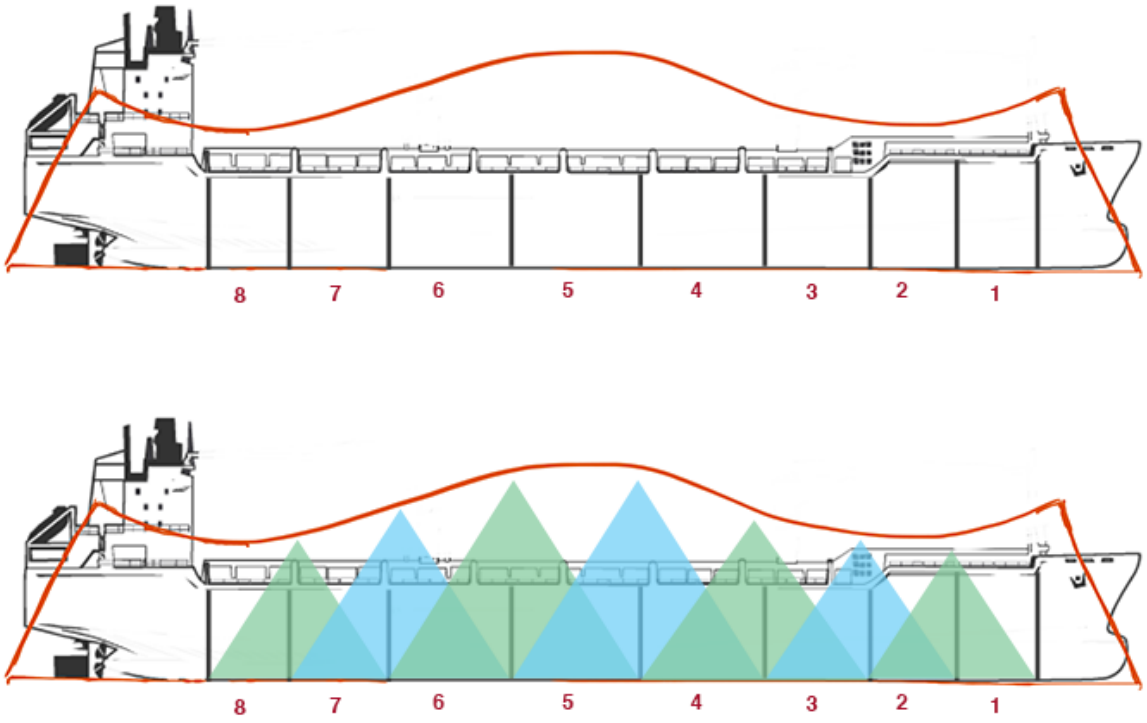


Figure 14.12 – Compliance with One Compartment Standard

The process can be repeated in a similar way if three compartment or four compartment ship is wanted.